

PokéTrader

Team 3

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Project Description

PokéTrader is a website for Pokémon enthusiasts who just want to collect Pokémon cards without having to spend too much money on singles and packs. The website offers a way to open virtual packs of real sets in the Pokémon Trading Card Game and store them in a virtual “binder” for their collection. Additionally, it also allows users to engage in trades for cards that they want.

The **primary goal** of PokéTrader is to provide an accessible, free experience to the Pokémon Trading Card Game. Pokémon products (booster packs, booster boxes, booster bundles, Elite Trainer Boxes (ETBs), collector boxes & tins, etc.) are relatively expensive for the average consumer to purchase.

The **major stakeholders** are primarily Pokémon fans, who will be the main target audience of this website application.

Our specific **innovation** is to provide more variety to enjoy the Pokémon Trading Card Game experience. Pokémon already has their Pokémon Trading Card Game Pocket, which is only available on mobile devices. We are providing a web-based application which can reach computer users who want to take part in Pokémon.

System Environment

System Structure

PokéTrader follows a three-tier architecture:



Client (Browser) Layer: Users will use a web browser to access the system. The interface is built using HTML, CSS, and JavaScript.

Application (Web) Layer: The application logic runs on Apache Tomcat using Java and JSP. This layer handles login and permission checks (i.e. for initiating trades & confirming trades).

Database Layer: MySQL stores all persistent data, including users, binder collection data, and trade history.

Hardware and Software used:

- OS: Windows/ macOS/ Linux
- Web Server: Apache Tomcat
- Database: MySQL Community Server
- Languages: Java, JSP, SQL, JavaScript, HTML, CSS

Functional Requirements

PokéTrader offers many different services for users. Every 12 hours, users can open one free Pokémon pack. Users can also check their virtual binder to see cards that they've already collected. To search for specific cards, users can type into the search bar to look for cards with specific characteristics (name, type, HP, stage of evolution, and weaknesses/resistances). Users can also push out requests for trades and those requests can be accepted by other users who are interested in trading for particular cards. During a trade, users can chat with each other to confirm the trade and then confirm the trade of cards, which then the PokéTrader application will swap the corresponding cards in each person's binder to trade.

User View

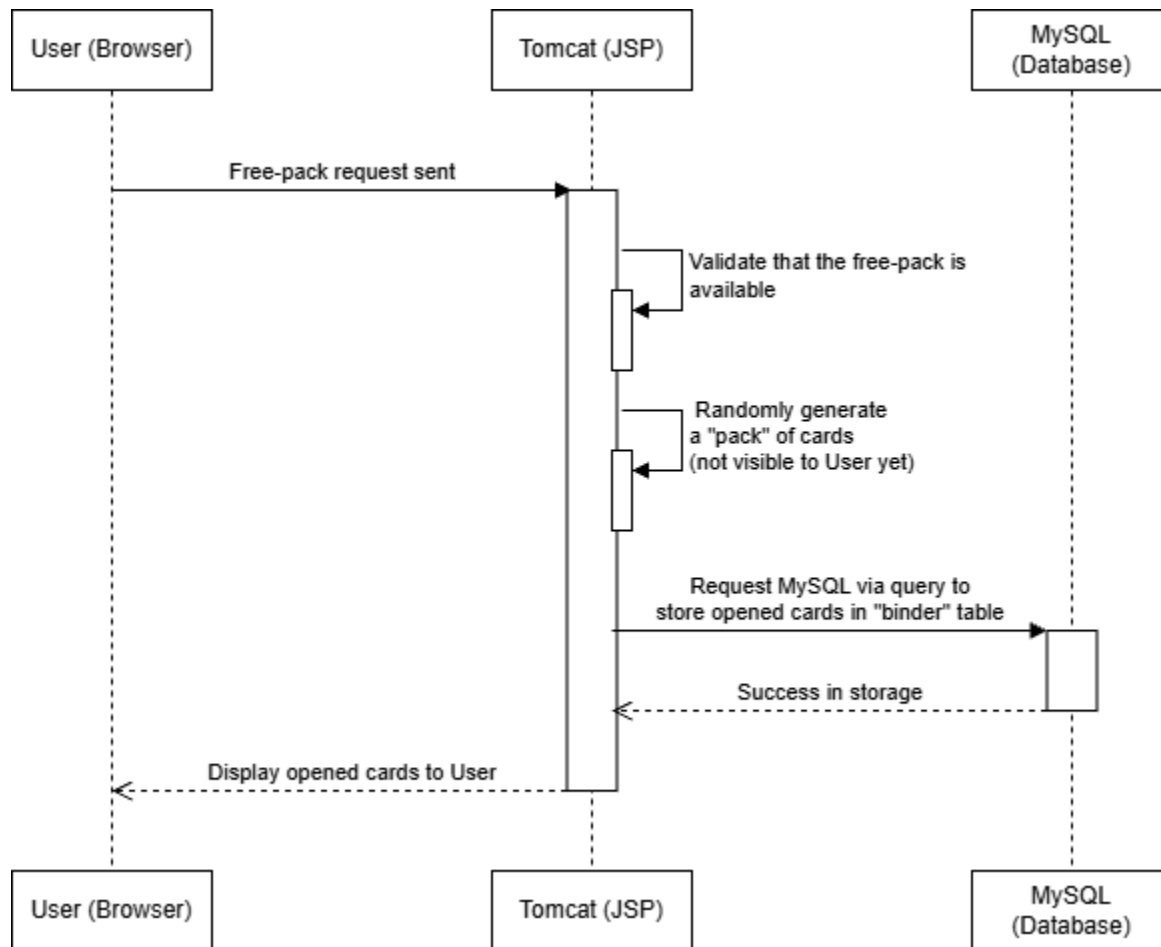
User Authentication

- Users must log in using their username and password in order to access their account.
- After login, the system creates a secure session for the user.
 - The first user using the account who is successfully logged into the system will be the only user who has access to the account. Other users who attempt logging into the same account will be denied access.

Pack-Opening Page

- Users will receive two free Pokémon packs of the latest set per day (i.e. one pack for every 12 hours). In order to redeem the free packs, users must navigate to the "Pack Opening Page" and click the "Open" button.
- On the top left of the page, there will be a message letting the user know when the next free pack will be available.

Example (Use-Case – User successfully opens their free pack of cards):



Binder Page

- Users can use the PokéTrader sorting system to display a subset of cards in the user's binder collection. This binder sorting system which will include the following:
 - A search bar
 - The user can type in the search bar to search (**case-insensitive**) these particular characteristics:
 - Name
 - e.g. "charizard" -> looks for cards named "Charizard"
 - "name = charizard" also works
 - Card-Type
 - e.g. "card-type = pokemon" -> looks for only Pokémon cards
 - e.g. "card-type = item" -> looks for only Item cards
 - e.g. "card-type = energy" -> looks for only Energy cards

- Pokémon-Type
 - e.g. “pokemon-type = grass” -> looks for only Pokémon cards that are Grass-types
- Expansion
 - e.g. “set = SVI” -> looks for cards only in the Scarlet and Violet set
 - “set = svi” also works
- Stage
 - e.g. “stage = basic” -> looks for Pokémon cards that are of the Basic evolution
- An “Advanced Options” button
 - Pulls up a selection of buttons to filter the following:
 - Card-Type
 - Pokémon
 - Item
 - Energy
 - Pokémon-Type
 - Grass, Fire, Water, Lightning, Psychic, Fighting, Darkness, Metal, Dragon, and Colorless
 - Stage
 - Note: This type is only relevant for Pokémon cards
 - Basic, Stage 1, and Stage 2
 - Legality
 - If the card is allowed to be played in a certain format
 - Formats:
 - Standard
 - Expanded
 - Gym Leader Challenge (GLC)
 - Pauper
 - Aceless
 - Expansion
 - Represented as 2-3 characters
 - e.g. Paldean Fates -> PAF
 - e.g. Diamond & Pearl -> DP
 - Rarity
 - Common (Black Circle ○)
 - Uncommon (Black Diamond ◇)
 - Rare (Black Star ★)
 - Double Rare (Double Black Star ★★)
 - Ultra Rare (Silver/Clear Stars ★★)

- Illustration Rare (Single Gold Star ★)
 - Special Illustration Rare (Double Gold Stars ★★)
 - Hyper Rare (Triple Gold Stars ★★★)
 - Promo (Black Star with “PROMO”)
- Users also have the option to flip through their binder collection by clicking either a “right arrow” button on the right side of the binder or a “left arrow” button on the left side of the binder.

Trade Page

- On this page, users can do the following:
 - Post a new trade request
 - The user will fill out a short form that details what is being offered as a trade (i.e. one or more cards) and can optionally put what cards they want to trade for.
 - Accept a trade request
 - Upon a user accepting this trade request, the trade request becomes closed to other users (except for the user who set up the request). The user has to wait in a “waiting room” until the other user joins the “waiting room”.
 - A notification is sent to the user who set up the request and they have the option to initiate the trade with the other user. Upon doing so, the user enters the waiting room.
 - Once both users (the user who accepted the request and the user who set up the request) are in the waiting room, they will go into a private chat room to discuss the offer.
 - The private chat room is set up to allow both users to see what cards are up for the trade and choose between clicking an “Accept Trade” button or a “Decline Trade” button.
 - If the trade is accepted, the respective cards will show up in the users binders. Within the binder, there will be a small icon indicating the information from the trade (date/time traded, user who you traded with, what you traded for, etc.).

Non-Functional Issues

Graphical User Interface:

- Interface should be simple to use
- Users should be able to go wherever they want in one to two clicks
- The GUI should include:
 - Clear, easy-to-read buttons
 - Color-coded tabs for each page

Security:

- In order to protect the user's account, the application will automatically log the user out of their account if one of the following things happen:
 - The user remains inactive on the application for more than 30 minutes.
 - The user quits the application.
 - The system crashes.

Access Control:

- Users
 - Can view their own binder
 - Can also sort through the binder to see subset of their collection
 - Can create trades and can accept trades

Performance:

- The system should be able to efficiently pull up the cards that the user wants to see in their binder.
 - Any trades made between two users should promptly reflect the changes in their respective binders.
- Notifications for waiting rooms should pop up on the requesting user's device quickly.

Reliability and Data Integrity:

- The system must prevent:
 - Multiple active users for one account (i.e. two [or more] users cannot use the same account)
 - A trade request that remains open to all other users even after one user has already accepted the request and is in the waiting room.